ONCOLOGY PATHOLOGY LABORATORY SERVICES

CYTOPATHOLOGY EVALUATION REPORT — FINE NEEDLE ASPIRATE (FNA) BIOPSY

Patient Reference ID: PT-842302 **Specimen:** Right Breast Upper Outer Quadrant

Clinical Indication: Palpable discrete breast nodule **Pathologist:** Dr. E. Bennett, MD, FCAP

COMPUTER-ASSISTED NUCLEAR MORPHOMETRY (WISCONSIN CRITERIA)

Digital image analysis of nuclear morphology from FNA aspirates. Thirty quantitative geometric metrics computed from digitized cell boundaries.

Nuclear Feature Metric	Clinical Quantitation	Nuclear Feature Metric	Clinical Quantitation
Radius Mean	17.99	Compactness Se	0.04904
Texture Mean	10.38	Concavity Se	0.05373
Perimeter Mean	122.8	Concave Points Se	0.01587
Area Mean	1001.0	Symmetry Se	0.03003
Smoothness Mean	0.1184	Fractal Dimension Se	0.006193
Compactness Mean	0.2776	Radius Worst	25.38
Concavity Mean	0.3001	Texture Worst	17.33
Concave Points Mean	0.1471	Perimeter Worst	184.6
Symmetry Mean	0.2419	Area Worst	2019.0
Fractal Dimension Mean	0.07871	Smoothness Worst	0.1622
Radius Se	1.095	Compactness Worst	0.6656
Texture Se	0.9053	Concavity Worst	0.7119
Perimeter Se	8.589	Concave Points Worst	0.2654
Area Se	153.4	Symmetry Worst	0.4601
Smoothness Se	0.006399	Fractal Dimension Worst	0.1189

DIAGNOSTIC PATHOLOGY IMPRESSION

Impression: Hyperchromatic, enlarged pleomorphic ductal epithelial cells exhibiting irregular nuclear contours and deep membrane invaginations. Morphometric dimensions substantiate high cytologic risk.